

Problem Solving (MATH 1000)

Summer 2018

Thursday, 05/24/2018

Quiz 1: Section 5.2, 5.3, 6.2, 6.4

Name (Print): _____

Score: _____ / 100

1. For each of the following items below, draw an example of a graph that satisfies the given properties:

(a)

10 points

 A graph with exactly 8 vertices and exactly 3 bridges.

(b)

10 points

 A graph with exactly 7 vertices and no bridges.

- (c) 10 points A graph with at least 10 vertices, at least 9 edges, and exactly 3 components.
2. 10 points Draw an example of a connected graph with exactly 9 vertices that is not simple but is regular and has two vertices with five degrees of separation.
3. 10 points Draw an example of a graph with up to 4 vertices that is not simple but is regular.

- [illegible]

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7. 10 points Draw any graph which has a Hamilton Circuit but not a Hamilton Path. However, if this is impossible, clearly and succinctly explain why.
8. 10 points Draw a connected, simple graph with six vertices which has no Euler Circuit, no Euler Path, no Hamilton Circuit, and no Hamilton Path. However, if this is impossible, clearly and succinctly explain why.